Designing and Building File-Folder Bridges as an Introduction to Engineering Design

Much of the material from:
COL Stephen Ressler, P.E., Ph.D.
Department of Civil & Mechanical Engineering
U.S. Military Academy, West Point

1

Why Study Bridges?

- Apply design process.
- Large structures need additional design steps, primarily for safety concerns.
  - Cannot build multiple real structures to test.
  - Cannot test until finished building.
- Designs like this need:
  - Computer simulation using physical principles to allow quick and inexpensive testing of alternate designs.
  - Real world data to put into computer model (e.g. strength of materials) to ensure accuracy.
  - Models to verify that computer simulation is correct.
  - Means of comparing the model data to the real structure.
- Learn more about forces.

2

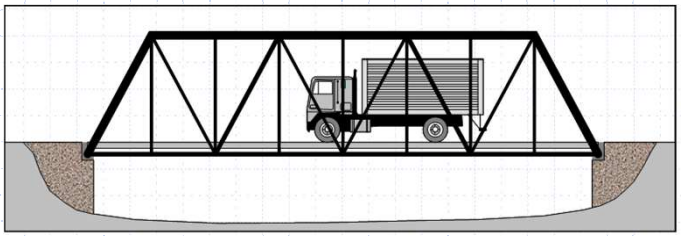
Why use file folders?

- Inexpensive.
- Easy to work with.
- Can make tubes, bars, and gussett plates that look and act like bridge structures.
- Behavior is predictable and compares surprisingly well to steel.
- Members are stronger than joints, like in real bridges.

3

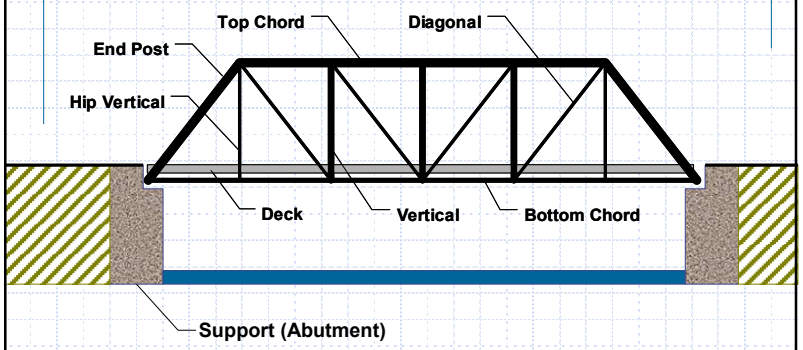
What is a Truss?

- A structure composed of members connected together to form a rigid framework.
- Usually composed of interconnected triangles.
- Members carry load in tension or compression.



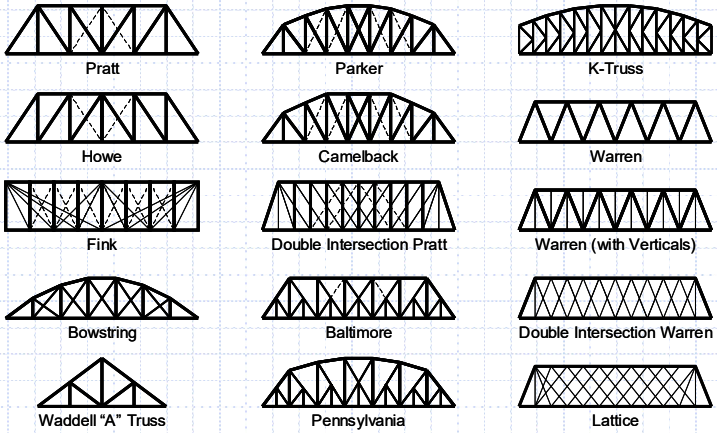
4

Component Parts



5

Standard Truss Configurations



6

Overall Plan

- Activity #1: Build a model of a truss bridge using file folders.
- Activity #2: Test the strength of structural members.
- Activity #3: Analyze and evaluate a truss. (Extra credit.)
- Activity #4: Design a truss bridge with a computer.
- Activity #5: Build a model truss bridge using your own design.

7

Read the Design Requirement

Design Requirement:

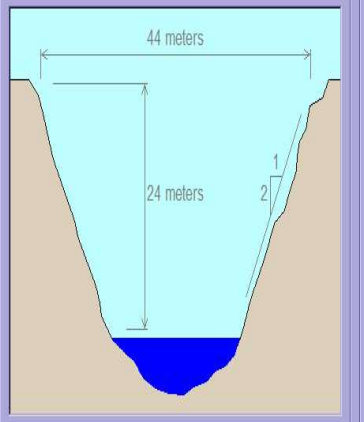
As a civil engineer working for the state Department of Transportation, you are responsible for designing a truss bridge to carry a four-lane highway across this river valley. Your objectives are:

(1) To ensure that the bridge can safely carry its own weight plus the weight of a standard truck loading.

(2) To keep the cost of the project as low as possible.

For the complete Design Specifications, click the Help button below.

Preview:



Design Tip:

The West Point Bridge Designer 2005 will help you satisfy these design requirements. The deck elevation and support configuration (which you will choose in Step 2) will be automatically set on the Drawing Board. WPBD will calculate the cost of your design as you work. WPBD also provides a simulated Load Test to check your bridge for structural safety. If your structure is not strong enough to carry the required loads, WPBD will tell you which members to strengthen.

8

Activity 1

- Activity #1: Build a model of a truss bridge using file folders.
  - Learn bridge terminology.
  - Learn construction techniques.
    - How to construct members.
    - How to follow plan.
    - How to turn into 3-D design.
  - Important when designing and building your own bridge.

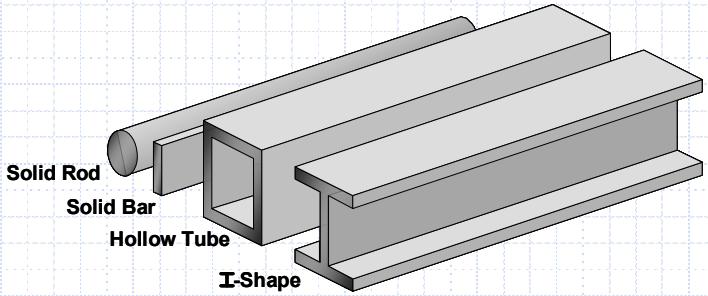
9

Activity 2

- Activity #2: Test the strength of structural members.
  - Make structural members out of cardboard.
    - Different size tubes.
    - Different size bars.
  - Test the strength under compression and tension. (Test to failure.)
  - Analyze and plot data.
  - Learn what affects strength.
  - Data will be used in scaling your final design from steel to paper to ensure your bridge can carry the load.

10

Types of Structural Members



These shapes are called cross-sections.

11

Testing Compressive Strength

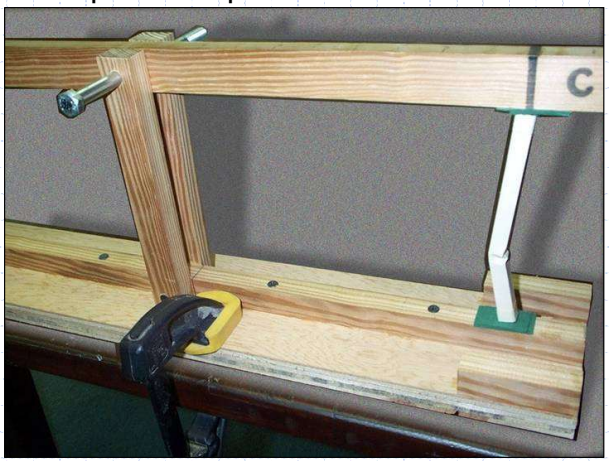
The test setup.



12

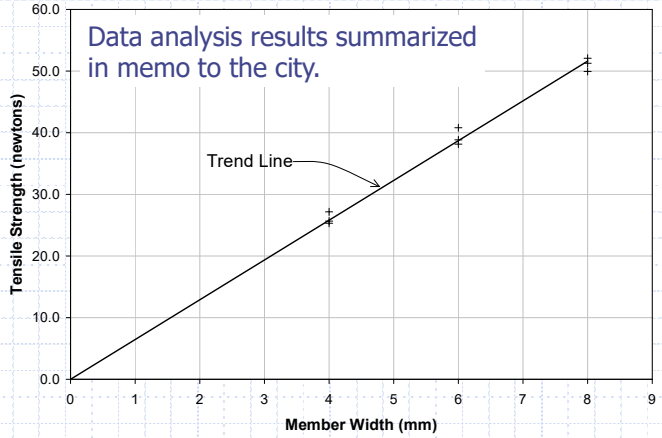
Testing Compressive Strength

A compression specimen at failure.



13

Graph the Results



14

Forces, Loads, & Reactions

- Force – A push or pull.
- Load – A force applied to a structure.

Self-weight of structure, weight of vehicles, pedestrians, snow, wind, etc.

- Reaction – A force developed at the support of a structure to keep that structure in equilibrium.

Forces are represented mathematically as VECTORS.

15

Activity 3: Analyze and Evaluate a Truss

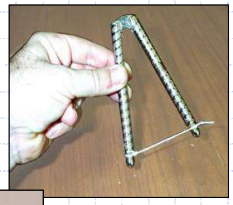
- May not do, but will at least discuss.
- Determine internal forces of compression and tension in the members of a bridge.
- Evaluate the safety of a bridge by comparing these forces to the strength of materials we found in Activity 2.
- The software will do this analysis for us when we design our own bridges.

16

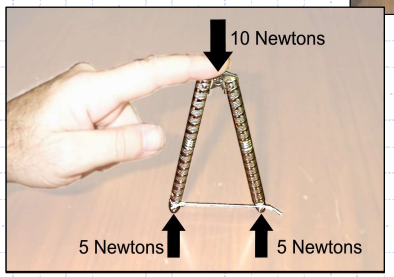
Equilibrium

Newton's First Law:

An object at rest will remain at rest, provided it is not acted upon by an unbalanced force.



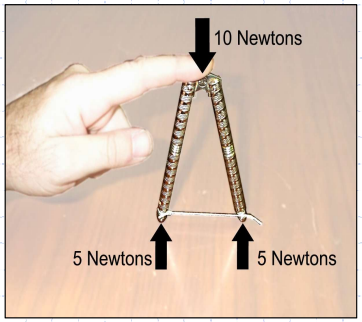
A Load...

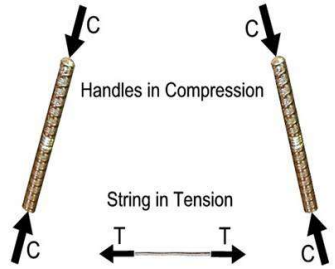


...and Reactions

17

Tension and Compression



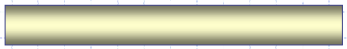


EXTERNAL FORCES and INTERNAL FORCES Must be in equilibrium with each other.

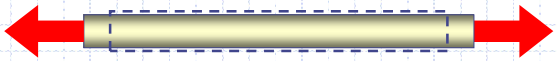
18

Tension and Compression

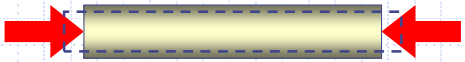
An unloaded member experiences no deformation



Tension causes a member to get longer



Compression causes a member to shorten



19

Activity 4: Design a truss bridge with a computer

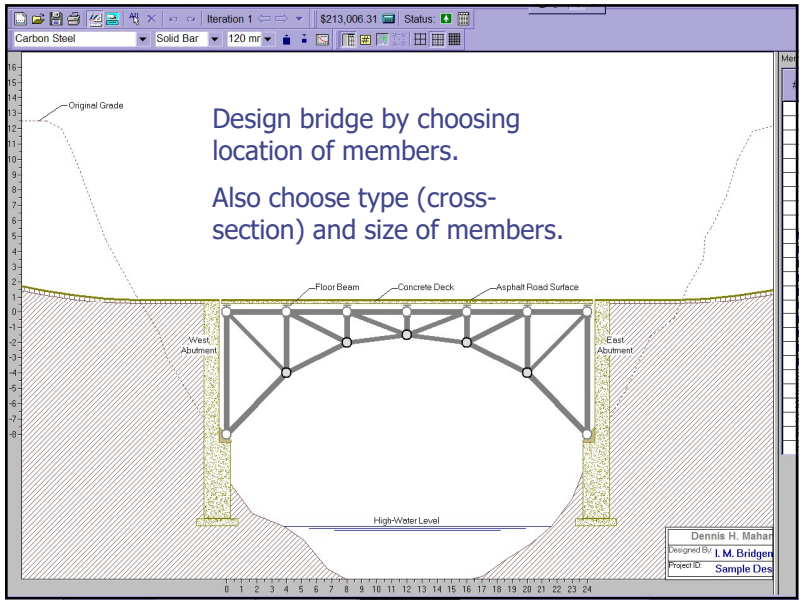
- Use West Point Bridge Designer Software.
  - Allows for quick and easy design of truss bridges.
  - Specific goal given. (Span, weight, cost, etc.)
  - Performs test to see if bridge fails.
  - Shows the forces in different members to allow identification of weak points.
- We will generate multiple successful designs.
- After comparing designs each team will choose the best.

20

The West Point Bridge Designer

- Look and feel of a standard CAD package.
- Easy to create a successful design.
- Hard to create a highly competitive design.
- Highly successful:
  - Over 150,000 copies downloaded since 2000.
  - Two major national software awards.
  - Formally endorsed as an educational tool by the American Society of Civil Engineers.
- Runs on Windows 95 (or later) PC.
- Can download to your own computer.

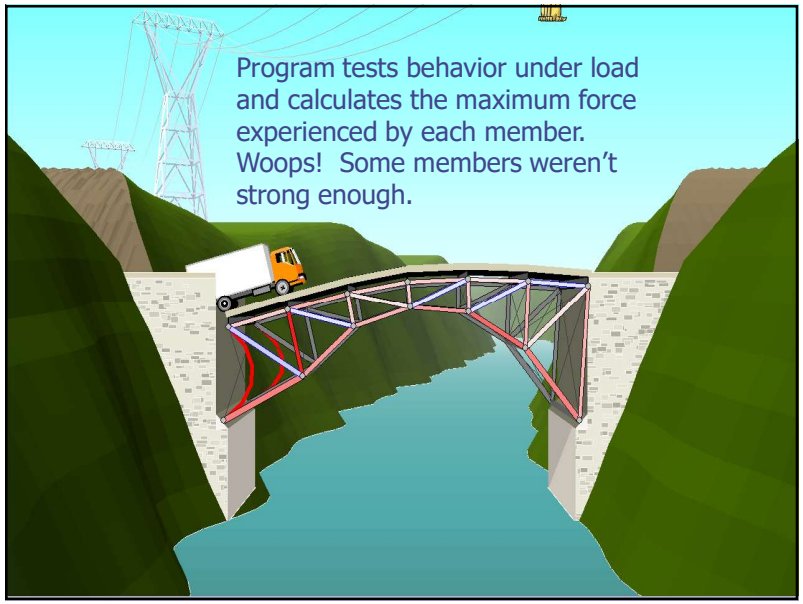
21



Design bridge by choosing location of members.

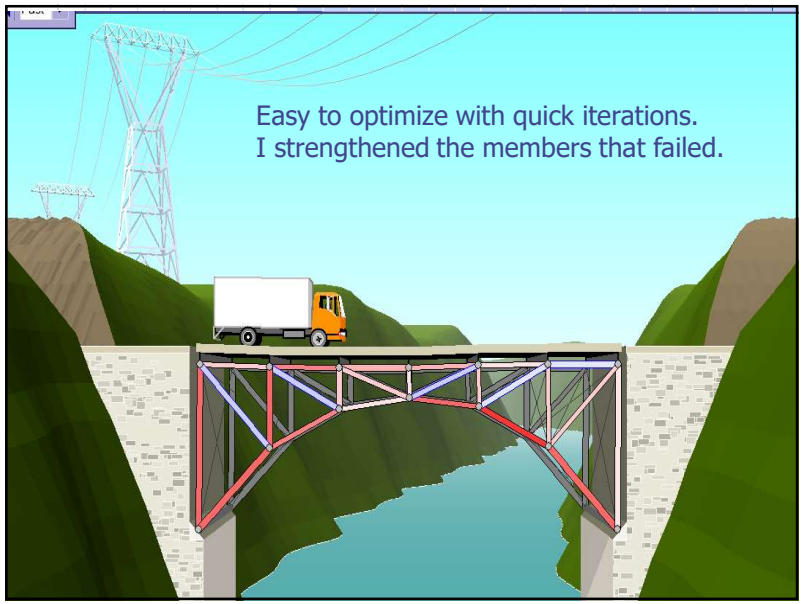
Also choose type (cross-section) and size of members.

22



Program tests behavior under load and calculates the maximum force experienced by each member. Woops! Some members weren't strong enough.

23



Easy to optimize with quick iterations. I strengthened the members that failed.

24

Structural Evaluation

- Our paper isn't exactly the same as steel.
- We need to determine if our model bridge can carry the weight before building!
- Is the internal member force less than the strength for each member?
- Calculate the Factor of Safety:

$$\text{Factor of Safety} = \frac{\text{Strength}}{\text{Internal Force}}$$

Activity 5: Design and build a model truss bridge

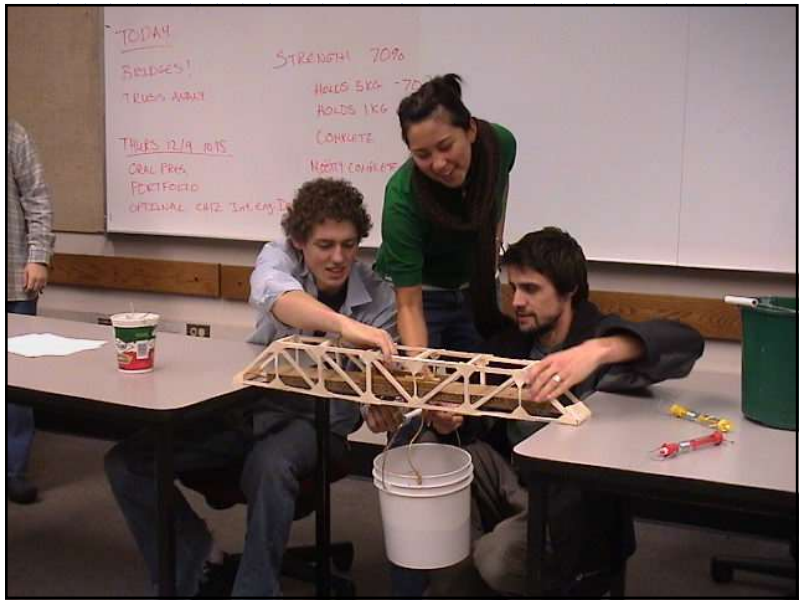
- Using our own designs from Activity 4 we will build bridges.
  - Same construction techniques as in Activity #1.
  - Use data from Activity #2 to ensure that individual members will not break under the applied force.
  - Data analysis needed to scale from a steel bridge with trucks driving across to a paper bridge with applied weight.
- Test bridges. (Will be done finals week. No final, but will need to be in class for the testing and party.)

Structural Design

- Design Requirements:
  - Span, loading, factor of safety

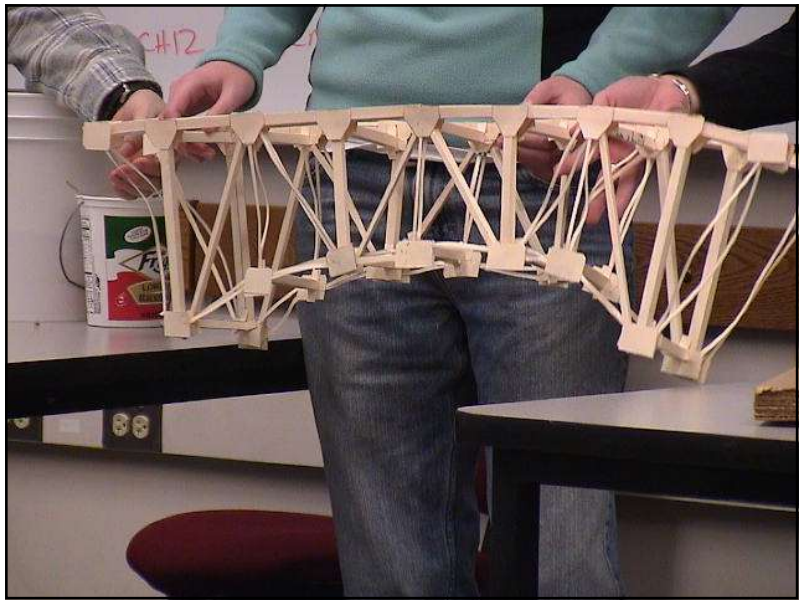


- Decide on truss configuration.
- Perform a structural analysis.
  - Reactions
  - Internal member forces
- Select member sizes based on required strength.
- Draw plans.
- Build the bridge.
- Test – *Can the bridge carry the required loading safely?*





29



30

Grading

	Output	Grade
Activity 1: Model	Bridge	5% (team)
Activity 2: Test members	Component Memo	25% (individual)
Activity 4: Design bridge.	Pugh evaluation	Graded with final output.
Activity 5: Build and test.	Bridge Web document	10% 30% (team)
	Participation	10% team 10% individ.
	Reflection	10% (individ.)

31

Summary

- File-folder bridges:
  - Accurate representation of real bridges
  - Vehicle for learning engineering design concepts.
  - Design based on authentic applications of math, science, and computer technology.
- The West Point Bridge Designer:
  - Experience the engineering design process.
  - Free!
- The West Point Bridge Design Contest:

West Point Bridge Design Contest
bridgecontest.usma.edu

32